

Group Project Overview

NYC Taxi Tipping & Weather Impact Pipeline

Overview:

The goal of our project is to build a distributed data pipeline to analyze the relationship between weather conditions and NYC taxi tipping behavior. We will ingest taxi trip data and historical weather data, store them in MongoDB, process them using Apache Spark, and develop machine learning models to predict tip amounts.

Dataset selection:

We have selected two complementary datasets to enable meaningful data fusion and analytics:

1. **Primary dataset (Dynamic): NYC Taxi & Limousine Commission (TLC) Trip Records.**
 - *Source:* NYC Open Data Socrata API.
 - *Format:* JSON/CSV (Structured).
 - *Volume:* High (Millions of rows).
 - *Key Fields:* `tpep_pickup_datetime`(yellow taxis), `trip_distance`, `fare_amount`, `tip_amount`(Credit cards only in this field), `passenger_count`, `PULocationID`, `DOLocationID`
- **Secondary dataset (Static): NYC Weather History**
 - *Source:* NOAA National Centers for Environmental Information - Central Park Station (USW00094728)
 - *Format:* CSV.
 - *Key Fields:* `DATE`, `PRCP` (Precipitation), `SNWD` (Snow Depth), `SNOW` (Snowfall), `TMAX` (Max Temp).

Project Rationale:

- **Realism:** Joining taxi trip transactions with weather context is the kind of “two-source” data work teams actually do in practice.
- **Scale:** NYC taxi data is huge, so it’s a good fit for showing distributed processing instead of toy examples. Processing it is standard and well-documented.
- **Clarity:** We can match trips to weather by date, which keeps the fusion simple and reduces messy edge cases.
- **Business Value:** We can test whether weather changes tipping patterns, which is a clear, practical question with business relevance.

Data Fusion Strategy

- **Join Key:** `Date` (YYYY-MM-DD).
- **Method:** We enrich the Taxi trip logs by joining them with the daily weather report. This allows us to analyze how rain or snow impacts trip duration and tipping behavior.

Analytics and Machine Learning Objectives

Data Engineering Pipeline

1. Ingest raw data from NYC API and NOAA into GCS via Apache Airflow DAGs
2. Load data into MongoDB for storage.
3. Perform an in-database aggregation (ETL) in MongoDB to fuse weather data with taxi trips based on the `date` key.

Analytics Goals

1. Use MongoDB Aggregation Framework to calculate "Average Tip Percentage" grouped by "Weather Condition".
2. Benchmark query performance: MongoDB Aggregation Framework vs. Spark SQL

Machine Learning Goal

1. **Objective:** Predict `tip_amount` for taxi trips using regression models
2. **Features:** `trip_distance`, `fare_amount`, `pickup_hour`, `PRCP` (Rain), `SNWD` (Snow).
3. **Models for Comparison:**
 - a. **Linear Regression:** To establish a linear baseline.
 - b. **Random Forest Regressor:** To capture non-linear behavioral changes (eg: potential tipping spikes during extreme weather).
4. **Evaluation Metrics:** RMSE, R^2 Score, MAE

Project Final Goal

By the end of the module, we will deliver a fully functional automated data pipeline that:

1. Uses Airflow to automatically pull and load data from more than one source on a schedule.
2. Joins the datasets and build aggregates in MongoDB to show the ETL step clearly.
3. Handles the larger-scale processing in Spark, using distributed SparkSQL.
4. Runs the same queries in MongoDB and Spark and compare runtimes, so we can show a fair performance comparison.
5. Trains ML models and report metrics, then summarize what the results mean in plain business terms.

Team Timeline and Milestones

Phase 1: Data Ingestion & MongoDB (Feb 7 – Feb 21)

- **Week 1 (Feb 8 - Feb 14):**
 - *Goal:* Set up Airflow DAGs.
 - *Task:* Write Python scripts to fetch API data and upload to GCS buckets (`raw-taxi`, `raw-weather`).
 - *Meeting:* Thursday, 12:00 PM (Code Review).
- **Week 2 (Feb 15 - Feb 21):**
 - *Goal:* MongoDB Integration (Deadline: Task 2, Feb 21st).
 - *Task:* Load GCS data into MongoDB collections. Write the `$lookup` aggregation to fuse weather data into the taxi collection.
 - *Meeting:* Thursday, 12:00 PM (Finalize Task 2 Submission).

Phase 2: Analytics & ML (Feb 22 – Mar 14)

- **Week 3 (Feb 22 - Feb 28):**
 - *Goal:* Spark Cluster Setup.
 - *Task:* specific Connect Spark to MongoDB. Load "Enriched Data" into Spark Dataframes.
 - *Meeting:* Thursday, 12:00 PM (Debugging connectivity).
- **Week 4 (Mar 1 - Mar 7):**
 - *Goal:* Model Training.
 - *Task:* Train Linear Regression and Random Forest models. Run performance benchmarks (SparkSQL vs. Mongo).
 - *Meeting:* Thursday, 12:00 PM (Review ML Results).
- **Week 5 (Mar 8 - Mar 14):**
 - *Goal:* Reporting (Deadline: Task 3 & Final Report, March 14th).
 - *Task:* Compile specific speed comparisons, ML metrics, and architecture diagrams into the final report.
 - *Meeting:* Thursday, 12:00 PM (Final Polish & Submit).

Conclusion

This project is a solid way for us to practice distributed data engineering end-to-end using real datasets and the tools from class. Our plan is realistic for the 5-week timeline, and our goals are clear enough to measure as we go. By March 14, 2026, we aim to deliver a working pipeline with data fusion, MongoDB aggregations, Spark-based processing, and an ML model, along with clear documentation of what we built and what we learned.

Phase 1

- Finalize the dataset/s
- Write up the summary of what analytics or what machine learning model to build
- Timeline - As a team, come up with goals for each week and a meeting schedule. Try to meet those goals for each week
- Keep that final goal in mind
- Discuss these goals with the team, and everyone agrees on the details

Write everything in a Word document, convert it into pdf and upload the pdf to Canvas.

Due Date: February 7, 2026, midnight

Phase 2

This is a mid-project checkpoint.

Task - 2 Airflow and MongoDB

- Load data into GCS (Google Cloud Storage)
- Import the data into MongoDB in collections (if you have multiple datasets, they should go in multiple collections.
- Create at least 1-2 new datasets from the original datasets based on some analytics, aggregation
- Store the aggregates in a separate collection in MongoDB
- Query the data in MongoDB (both original and new aggregate data) using MongoDB queries and/or SparkSQL.

This is a good time to evaluate and assess based on the initial task list and timeline, where you stand.

Submit any files (code, scripts, DAGs, documentation) as a single zip file to Canvas. There are no set expectations or guidelines. This is just a rough check-in.

Due Date: Feb 21 at 11:59pm

Phase 3

Task - 3 Airflow MongoDB + SparkSQL

- Create Dataframes from the data from MongoDB Atlas
- Run SparkSQL queries over the data frames
- Build machine learning algorithms to derive analytics on top of the original and aggregated datasets

Due: March 14, 2026 Midnight

Task Description

Distributed Data Systems - Group Project

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Group Project

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Your Team Your Project Your Success

Five to Six people per team.

Canvas to create the groups automatically.

Everyone needs to contribute and work!

Peer-feedback is important.

3

Build an automated data pipeline

Use Airflow to connect different systems like MongoDB, Spark Cluster, GS storage

Understand Map-Reduce or aggregation pipeline in MongoDB

Do data exchange between MongoDB and Spark

*** Use your data science skills to implement a machine learning model of your choice

Group Project Objective

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Group Project Goals

Gather a suitable dataset (preferred if there are more than 1 dataset for more realistic data processing scenario).

More use cases for data aggregation and building ML models.

Consider your final goal:

Make sure that the goal is reachable within the 7 weeks.

Detailed project report (template soon available on canvas)

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Dataset selection

Some Data Sources

- Your own data (the best) : e.g. Apple SensorLog, etc.
- data.gov: <https://www.data.gov/>
- Twitter API: <https://developer.twitter.com/en/docs/twitter-api>
- Meetup API: <https://www.meetup.com/api/guide/>
- Football Data API: <https://www.football-data.org/>
- US Patent and Trademark Office API:
<https://www.uspto.gov/learning-andresources/open-data-and-mobility>
- Many more!! - <https://github.com/public-apis/public-apis>

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Things to consider

- Data fusion from multiple data sources
- Conceptualize what data processing/aggregations/new analytics you want to define
- Develop and/or Apply novel data processing / ML algorithms
- Compare results of different ML algorithms

- Compare results from different machine specs- Costs/Speed/DataSize

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Task - 1 Datasets + Project Plan

- Finalize the dataset/s
- Write up the summary of what analytics or what machine learning model to build
- Timeline - As a team, come up with goals for each week and meeting schedule. Try to meet those goals for each week
- Keep that final goal in mind
- Discuss these goals with the team and everyone agrees on the details

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Due: February 7, 2026 Midnight

Task - 2 Airflow and MongoDB

- Load data into GCS (Google Cloud Storage)
- Import the data into MongoDB in collection
- Create at least 1-2 new datasets from the original datasets based on some analytics, aggregation
- Store the aggregates in a separate collection in MongoDB on MongoDB Atlas
- Query the data in MongoDB (both original and new aggregate data) using MongoDB queries.

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Due: February 21, 2026 Midnight

Task -3 Airflow MongoDB + SparkSQL

- Create Dataframes from the data from MongoDB Atlas
- Run SparkSQL queries over the data frames
- Build machine learning algorithms to derive analytics on top of the original and aggregated datasets

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Due: March 14, 2026 Midnight

Submission (Blog Post/short paper style)

- Your dataset/s and goals
- Rational behind the goals and the choice of the dataset/s
- Overview of your data engineering pipeline
- Preprocessing goals, algorithms and time efficiency
- Explain the different analytics and queries you ran over the datasets in MongoDB and on Dataframes in Spark
- Compare and contrast the time of the queries over both the version of the datasets
- Lessons learned
- Conclusion